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**ACADEMIC AND SCIENTIFIC RESEARCH**

**Pedestrian Trajectory Prediction**

From Synthetic Data to Real-World Scenarios using LSTM and PSO  
Optimization

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## ABSTRACT

This report investigates pedestrian trajectory prediction with a progressive experimental strategy: we first validate the pipeline on synthetic trajectories to isolate modeling issues, then transfer the approach to real-world pedestrian data (ETH/UCY), where noise and irregular motion increase difficulty. We evaluate LSTM-based forecasting under multiple trajectory representations (absolute, normalized, offset) and across several horizons. To reduce manual tuning, Particle Swarm Optimization (PSO) is used to optimize key hyperparameters. Results highlight the expected performance gap between synthetic and real settings, show that offset-based modeling improves stability, and confirm that classical baselines (e.g., Constant Velocity) remain strong for long-horizon prediction. We conclude with limitations and planned improvements.

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# 1 INTRODUCTION

Pedestrian trajectory prediction aims to forecast future two-dimensional positions of individuals based on a sequence of past observations. This problem is central to many safety-critical applications such as autonomous driving, intelligent surveillance, and human–robot interaction, where anticipating pedestrian motion is essential for safe and socially compliant decision-making.

Despite notable progress, accurate trajectory prediction remains challenging, particularly over long prediction horizons. Pedestrian motion is inherently uncertain and affected by multiple factors, including individual intent, environmental constraints, and interactions with other agents. These factors lead to highly non-linear and diverse motion patterns, causing prediction errors to accumulate rapidly as the horizon increases.

Recent research has increasingly focused on complex deep learning architectures, such as attention-based or interaction-aware models, to capture social behaviors and contextual information. While effective, these approaches often come at the cost of increased complexity, reduced interpretability, and high data and tuning requirements.

In contrast, this work adopts a deliberately simple and controlled approach. We investigate the capabilities of a standard LSTM-based model when combined with appropriate trajectory representations and systematic hyperparameter optimization. Our methodology follows a progressive strategy: we first validate the modeling pipeline on synthetic trajectory data under controlled conditions, then transfer the same framework to real-world pedestrian trajectories from the ETH/UCY benchmark, where noise and irregular motion significantly increase task difficulty.

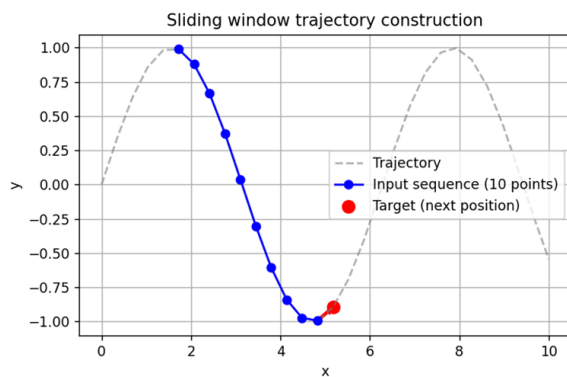
To further improve robustness and reproducibility, Particle Swarm Optimization (PSO) is employed to automatically tune key hyperparameters such as learning rate, hidden dimension, and dropout. This allows a systematic exploration of the parameter space and provides insight into stable configurations for pedestrian trajectory prediction.

## 2 RELATED WORK

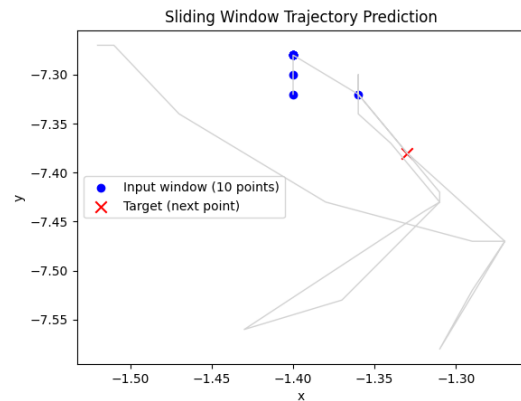
### 2.1 Datasets

| Dataset   | Type       | Scenes / Patterns              | Sampling           | Notes                  |
|-----------|------------|--------------------------------|--------------------|------------------------|
| Synthetic | Artificial | Linear/turning/CV              | $\Delta t = \dots$ | Noise-free validation  |
| ETH/UCY   | Real       | ETH, Hotel, Univ, Zara1, Zara2 | $\Delta t = \dots$ | Crowded outdoor scenes |

Table 1: Datasets used in this project. Replace  $\Delta t$  and details with your values.



(a) Synthetic trajectory with sliding window construction. The input sequence (10 observed positions) is shown in blue, while the prediction target corresponds to the next position in red.



(b) Sliding window construction on real ETH/UCY pedestrian trajectories. The same windowing strategy is applied as for synthetic data.

Figure 1: Illustration of the sliding window trajectory formulation used for both synthetic and real-world datasets.

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## 2.2 State-of-the-Art Baselines

| Family         | Method               | Key idea                                  |
|----------------|----------------------|-------------------------------------------|
| Classical      | Last Position        | Predict no motion                         |
| Classical      | Constant Velocity    | Linear extrapolation from recent velocity |
| Learning-based | LSTM/RNN             | Temporal modeling from sequences          |
| Advanced       | Social/Attention/GNN | Model interactions and context            |

Table 2: Trajectory prediction families and common baselines.

### 3 PROPOSED SYSTEM

#### 3.1 Pipeline Overview (insert your diagram here)

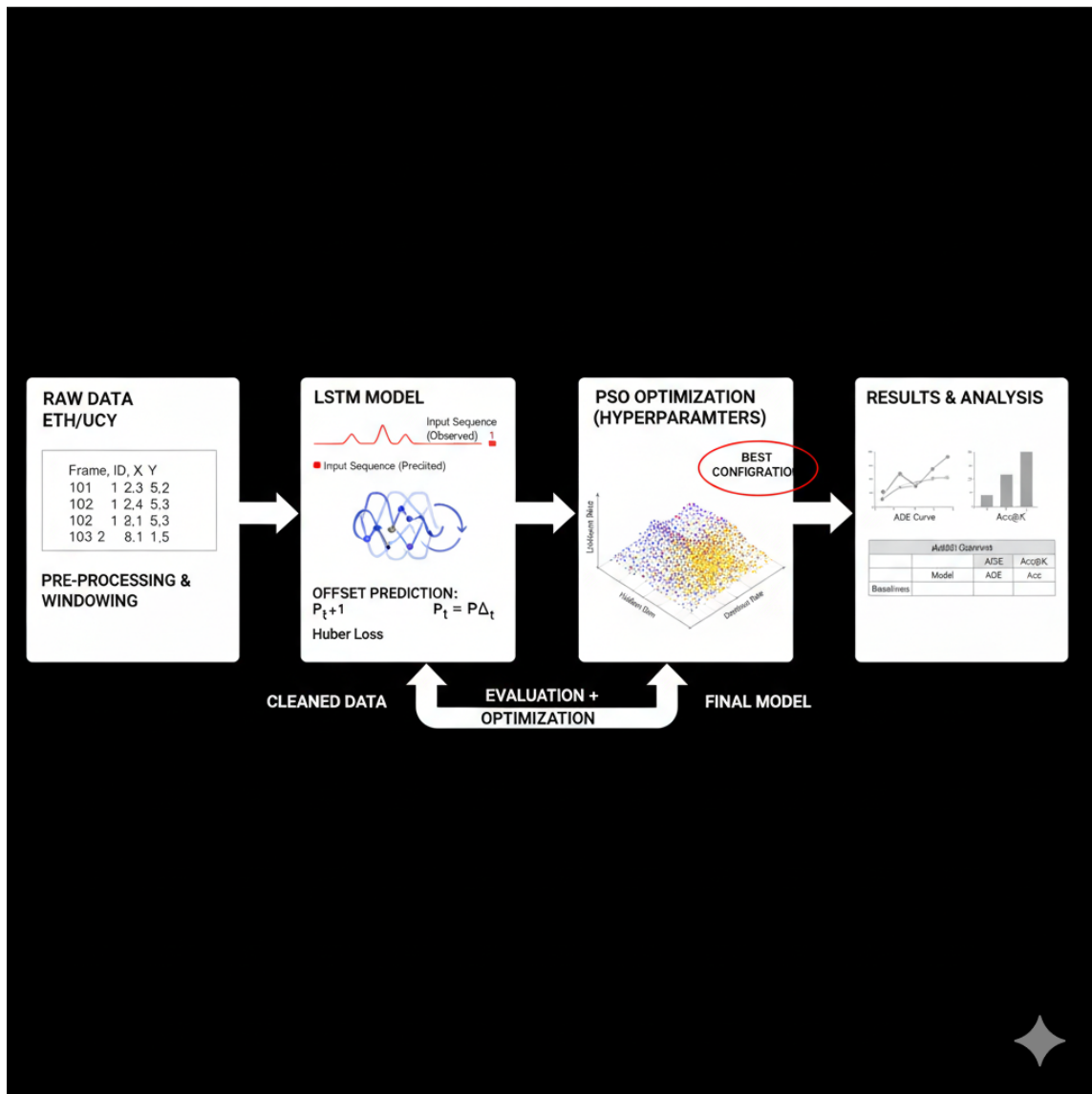


Figure 2: Proposed pipeline: preprocessing, windowing, LSTM prediction, evaluation.

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## 3.2 Trajectory Windowing

| Parameter                        | Value              |
|----------------------------------|--------------------|
| Observation length ( $T_{obs}$ ) | 10 timesteps       |
| Prediction horizons ( $H$ )      | 8, 12, 20, 30      |
| Input dimension                  | $T_{obs} \times 2$ |
| Output                           | 2D position(s)     |

Table 3: Sliding-window formulation.

## 3.3 Representations and Loss

| Variant    | Target                                    | Motivation                             |
|------------|-------------------------------------------|----------------------------------------|
| Absolute   | $(x_{t+H}, y_{t+H})$                      | Direct but position-dependent          |
| Normalized | normalized coords                         | Reduce scale variance                  |
| Offset     | $\Delta = (x_{t+H} - x_t, y_{t+H} - y_t)$ | Easier learning, translation-invariant |

Table 4: Modeling variants compared in this work.

| Loss  | Comment                                           |
|-------|---------------------------------------------------|
| MSE   | Sensitive to outliers (noisy real trajectories)   |
| Huber | Robust; smooth near zero, linear for large errors |

Table 5: Loss functions used for training.



## 4 EXPERIMENTS

### 4.1 Metrics

| Metric     | Definition                 | Interpretation   |
|------------|----------------------------|------------------|
| ADE        | avg. distance over horizon | lower is better  |
| MSE        | squared error              | lower is better  |
| Accuracy@k | $1(d \leq k)$              | higher is better |

Table 6: Evaluation metrics.

### 4.2 Synthetic vs Real Comparison

| Dataset        | ADE ↓ | Acc@1.0 ↑ | Acc@2.0 ↑ |
|----------------|-------|-----------|-----------|
| Synthetic      | ...   | ...       | ...       |
| Real (ETH/UCY) | ...   | ...       | ...       |

Table 7: Performance gap: synthetic vs real. Fill with your results.

### 4.3 Baselines (Real Data)

| Model              | ADE ↓ | Acc@1.0 ↑ | Acc@2.0 ↑ | Acc@4.0 ↑ |
|--------------------|-------|-----------|-----------|-----------|
| Last Position      | ...   | ...       | ...       | ...       |
| Constant Velocity  | ...   | ...       | ...       | ...       |
| LSTM Absolute      | ...   | ...       | ...       | ...       |
| LSTM Normalized    | ...   | ...       | ...       | ...       |
| <b>LSTM Offset</b> | ...   | ...       | ...       | ...       |

Table 8: Baseline comparison on real-world trajectories.

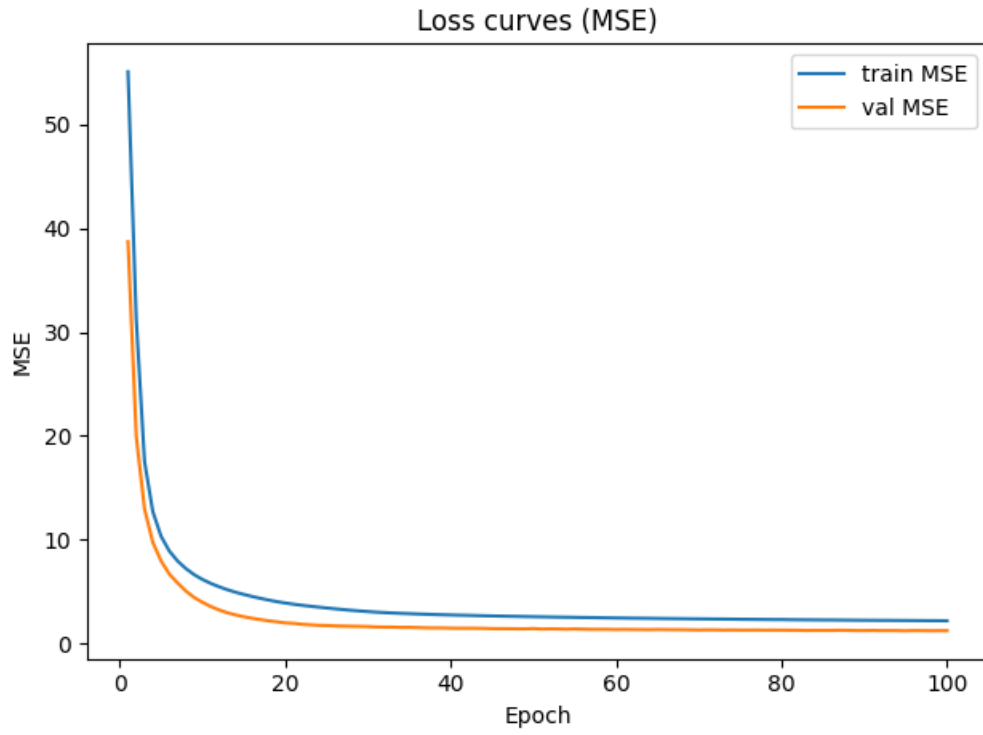


Figure 3: Training/validation loss curves.

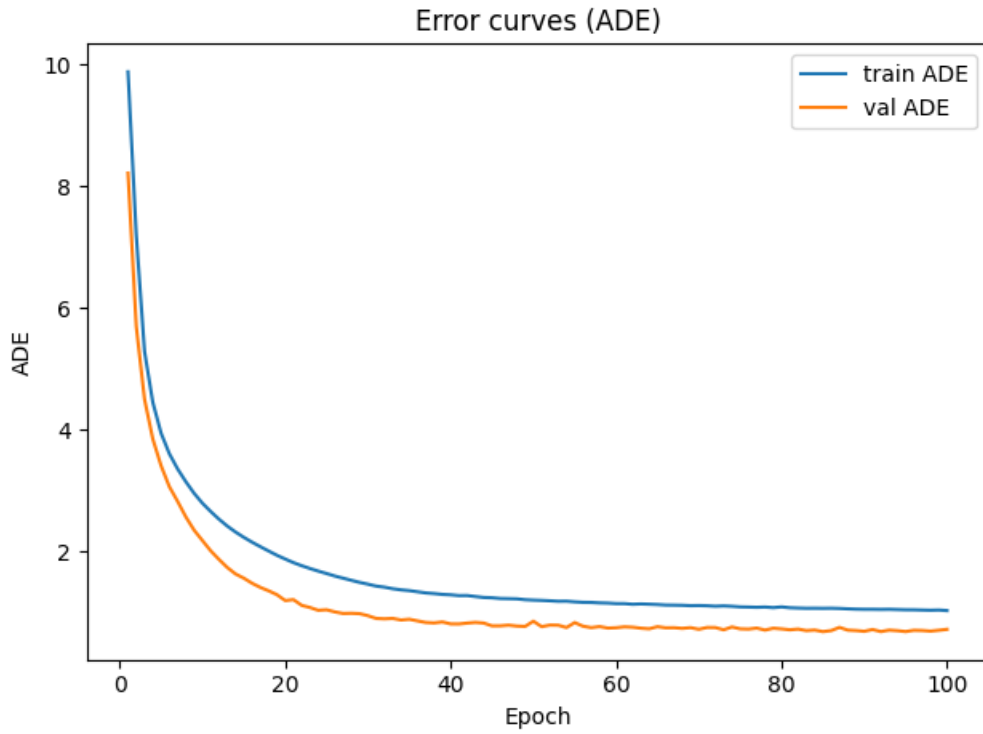


Figure 4: ADE evolution across epochs.

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## 4.4 Long-Horizon Analysis

| Horizon $H$ | Offset LSTM ADE ↓ | CV ADE ↓ | Winner |
|-------------|-------------------|----------|--------|
| 8           | ...               | ...      | ...    |
| 12          | ...               | ...      | ...    |
| 20          | ...               | ...      | ...    |
| 30          | ...               | ...      | ...    |

Table 9: Long-horizon robustness comparison.

## 4.5 PSO Optimization

| Hyperparameter | Search range         | Best value (PSO) |
|----------------|----------------------|------------------|
| Learning rate  | $[10^{-5}, 10^{-3}]$ | ...              |
| Hidden dim     | $[16, 128]$          | ...              |
| Dropout        | $[0, 0.3]$           | ...              |

Table 10: PSO search space and best configuration (fill values).

| Configuration | Validation ADE ↓ |
|---------------|------------------|
| Manual tuning | ...              |
| PSO optimized | ...              |

Table 11: Impact of PSO optimization.

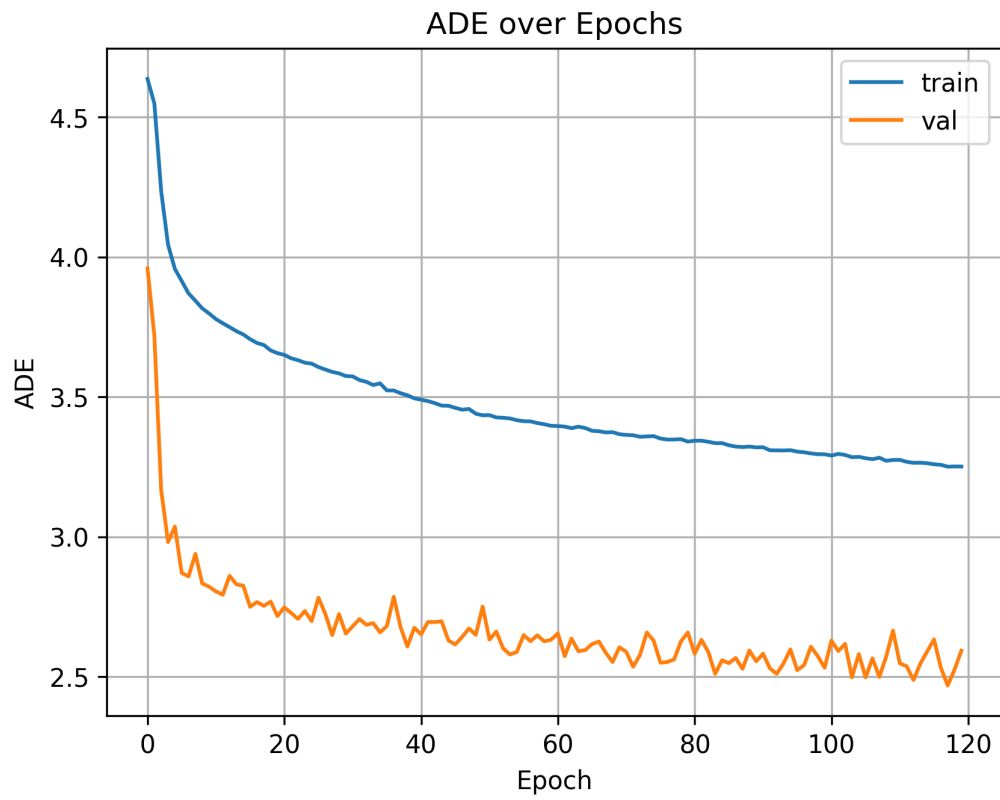


Figure 5: PSO convergence curve (insert your run image).

## 5 CONCLUSIONS

| Item         | Summary                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Key findings | Synthetic validation helps debugging; offset formulation improves stability; PSO reduces manual tuning; CV baseline remains strong at long horizons.       |
| Limitations  | No interaction modeling; error increases sharply for large horizons; limited generalization across scenes if training is scene-specific.                   |
| Future work  | Interaction-aware models; probabilistic prediction; multi-step decoding; curriculum learning from synthetic to real; robustness evaluation on more scenes. |

Table 12: Conclusion summary (findings, limitations, future work).